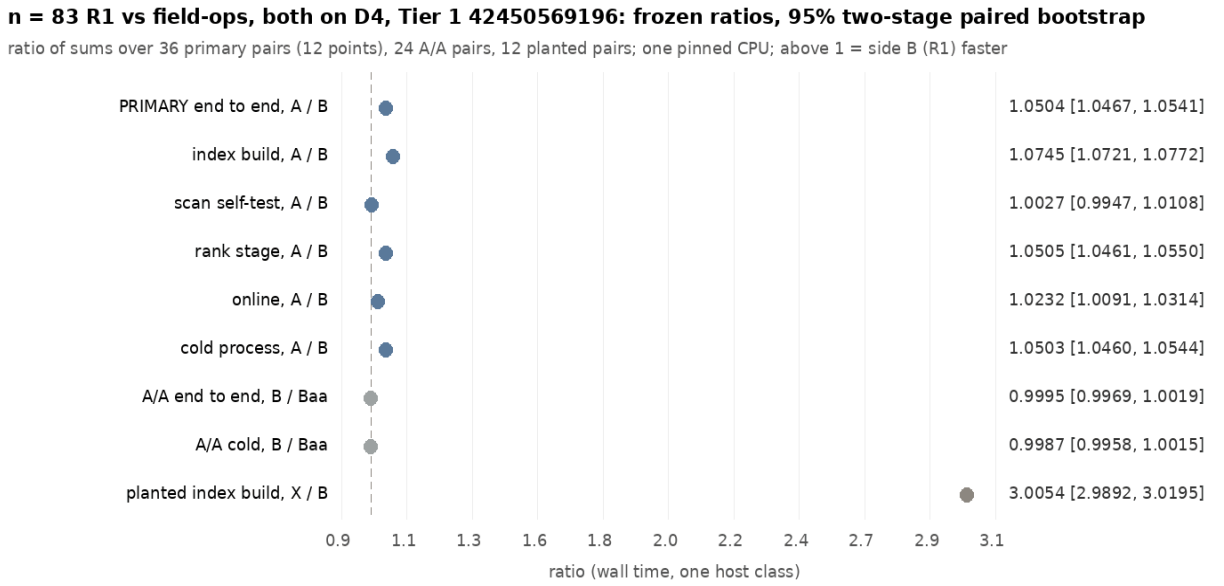
n = 83: field-ops-2 (R1) against field-ops, both on the D4 base

Status (2026-11-08): result in. Verdict: improvement; every gate and control passes and the run is valid `[pcc:unverified]`. R = E2E(field-ops) / E2E(R1), both on D4, = **1.0504 [1.0467, 1.0541]** over 36 alternating pairs, run 42450569196, Tier 1, instance scope. The design sections below are the frozen text (commit `bca22e9f9`, before any timing on the Tier 1 runner); the result section is added with the archived receipt.

Result

Run 42450569196 (workflow `n83-fo2d4.yml`, push trigger, head `bca22e9f9`, attempt 1, conclusion success) on one GitHub-hosted `ubuntu-24.04` runner (`GitHub Actions 1000203805`, **Intel Xeon Platinum 8573C**, execution CPU 0; the k4c400 run was on a Xeon Platinum 8370C). All 278 scheduled rows ran and are valid, none was skipped by the time guard. The receipt is in `run_github_42450569196/` (`panel/runs.jsonl`, 1.4 MB, in the artifact store).

Verdict: improvement `[pcc:unverified]`: R = sum E2E(A) / sum E2E(B) = **1.0504 [1.0467, 1.0541]** over 36 alternating pairs (12 points x 3 pairs; two-stage paired bootstrap, 10,000 draws), A = field-ops producer `ic_online_fieldops_r770`, B = R1 `ic_online_fieldops2_r770`, both on D4. Every gate passes (`verify-stdout.json`): one scalar, relation, probe count and index and layout digest set per point across both binaries, equal to D4's frozen identity; Q0 equal for both binaries; identical rank relations, probe vectors, root calls and counters per seed across kD4FO, kD4R1 and kD4R1aa; rank tables equal to the frozen table; pipelines and binaries as the arms name them; placement and host identity unchanged. `verify_receipt.py` and `band_model.py` re-run on a full copy of the receipt directory reproduce `panel/result.json`, `panel/result_table.md` and `band/band_model.json` byte for byte.



stage (A = field-ops, B = R1, both D4)	side A mean (ms)	side B mean (ms)	A / B [95%]
index build (cold process)	2,107	1,961	1.0745 [1.0721, 1.0772]
scan self-test	209	208	1.0027 [0.9947, 1.0108]
rank stage (rank process)	1,878	1,787	1.0505 [1.0461, 1.0550]
online, one target	695	679	1.0232 [1.0091, 1.0314]
cold process	3,404	3,241	1.0503 [1.0460, 1.0544]
end to end (rank + cold)	5,281	5,028	**1.0504 [1.0467, 1.0541]**

The stage rows are stage secondaries and never decide. The index build carries most of the saving (7.5%), the rank stage 5.1%; the self-test, which runs no field products, is unchanged, as in the field-ops-2 run. Process CPU time A / B 1.0430 [1.0403, 1.0463]; peak RSS equal (0.99997).

- **A/A** (B against Baa, 24 pairs): end to end 0.9995 [0.9969, 1.0019], cold 0.9987 [0.9958, 1.0015] (both inside [0.97, 1.03]).
- **Planted:** cD4R1X / cD4R1 index build 3.0054 [2.9892, 3.0195] (lower bound above 1), layout digests unchanged.
- **Probe law** (estimate, band [0.80, 1.20], 144 attempts each): kD4FO 0.9916 [0.7738, 1.2394], kD4R1 0.9916 (the same work on both binaries, as gated).
- **Order and spread:** A first 1.0527, B first 1.0492; pair positions 1.0504, 1.0486, 1.0521; per-point ratios 1.037 to 1.060 (12 of 12 above 1). Geometric mean of the per-pair ratios 1.0506 [1.0469, 1.0541].

- **Against the power check:** the design had power 0.998 at a true 1.010 and the estimate is 1.050, far above the minimum effect. Observed per-side E2E CV 0.209 (A) and 0.210 (B): large, but the two sides do identical work per pair, so the ratio's noise is small (A/A pair noise in `analysis/power.json`).

Expected effect was about 2-3% `[pcc:unverified]`; the measured 5.0% is larger than that expectation and than R1's 2.56% on the M38 base: on D4 the index build (7.5% faster here against 5.0% on M38) is a larger share and the rank stage gains 5.1% against 0.6% there. Ratios from different runs do not compose, so this is its own measurement, on another runner CPU class.

Distance to rho

In this run's matched rT770 rho (12 processes, mean cold 2,461.5 ms, step 254.81 ns) the end-to-end cold gap is **2.1455 for side A (field-ops on D4) and 2.0426 for side B (R1 on D4)** (point estimates, no interval computed for the gaps); S end to end cold is 0.2239 for A and **0.2132 for B** in rT770 steps `[pcc:unverified]`. The k4c400 run read 2.0485 for the field-ops producer on D4 on another runner (rT770 step 275.68 ns); gaps do not compose across runs (the matched rho's cold mean is a mean of random-length processes and moves with panel and host). Wall time, one host class, factor-base construction charged to neither side, so cold S is a lower bound.

Section 5a (priced native work; a model band, not a counted band)

`band_model.py` (frozen) models the IC units from the archived counted run 39882571652; neither producer here emits operation counts, so this run has no counted band and none was run, in CI or locally. Both sides do identical group-operation work (gated), so the model gives A and B the same units and cannot see a field-arithmetic change: at 0, 0.1 and 1 GAE per unpriced native operation the gap to rT is 9.844e-05, 0.7217 [0.4782, 1.177], 2.875 [1.905, 4.69] for both sides; the order against rho is unresolved (it changes by w = 1). Per the coordinator's note on PR #1448 (opcounts2, merged; not re-checked here) `[pcc:unverified]`, this model under-states the unpriced IC operations by about 1.4x, so the in-band gaps above are probably low. A counted band needs the counted producer of

1448 run on this design; not done.

Limits of this result

- Wall time on one GitHub-hosted runner (Intel Xeon Platinum 8573C), one pinned CPU, Tier 1 instance scope, no host-wide exclusivity; not replayed on another host class (the k4c400 run was on an 8370C, field-ops and field-ops-2 on an AMD EPYC 9V74).
- 36 pairs on 12 points; the narrowed panel scan is disclosed above.
- Factor-base construction is charged to neither side, as in every n = 83 row.

Files (result)

`run_github_42450569196/` (receipt, `band/`), `figures/make_result_figure.py`, `figures/result.svg`, `figures/result.png`, `figures/make_report_pdf.py`, this report as `README.pdf`.

Question

The headline row is the field-ops producer (`ic_online_fieldops_r770`) on the D4 closure base (K' = 4, 400 orbits; `koblitz-n83-k4c400-20261107`). The field-ops-2 producer R1 (`ic_online_fieldops2_r770`, a constant reduce shift at n = 83; `koblitz-n83-fieldops2-20261107`) was measured only on the M38 base (1.0256 [1.0236, 1.0278]) `[pcc:unverified]`. **Is the end-to-end cold cost of R1 on D4 lower than that of the field-ops producer on D4?** Both sides run the same header, table, points and rank seeds; only the binary differs. Ratios from different runs do not compose, and D4 shifts the stage mix, so the stacked effect is measured. Expected effect: about 2-3% `[pcc:unverified]` (an expectation).

Design (frozen before any timing)

- **Sides:** A = `kD4FO` + `cD4FO` (field-ops binary, built from the field-ops-2 experiment's `online/` crate); B = `kD4R1` + `cD4R1` (R1, its `online-b/` crate); A/A = B against `kD4R1aa` + `cD4R1aa`; planted = `cD4R1` against `cD4R1X` (`lift_plant`, index build); matched rho `rT`. `+pclmulqdq` on both, one toolchain (1.99.0 in CI).
- **Panel:** 12 new held-out points (`inputs/cold_targets.json`, role `fo2d4-cold-v1`), disjoint from every earlier n = 83 panel including k4c400's (`inputs/panel_disjointness.json`, 0 hits; narrowed scan: origin/main in full and the 92 origin branches whose name contains `n83`, as in k4c400, disclosed).
- **Rank seeds:** 2601-2636, new on n = 83 (`inputs/seed_disjointness.json`), outside 2201-2236, 2301-2360 and 2501-2560.
- **Pairs per point:** 3 alternating primary pairs (same point and rank seed, seeded ABAB/BABA), 2 A/A pairs, 1 planted pair, 1 rho process; 278 runs, about 24 minutes modelled.
- **Primary:** R = sum E2E(A) / sum E2E(B) over 36 pairs, E2E = rank stage + cold process; improvement iff the 95% two-stage paired bootstrap lower bound exceeds 1 (10,000 draws).
- **Primary choice (frozen before data): end to end, not cold only.** The noise of one A/B pair comes from the k4c400 A/A pairs (identical work): sd 0.0103 in log for E2E, 0.0121 for cold, so E2E is no noisier. At 12 x 3 pairs E2E resolves a 1.010 effect with power 0.998 and 1.005 with 0.67 (cold only 0.997 and 0.56; `analysis/power.json`). A cold-only schedule would save about 12 minutes and would measure another quantity (R1 also touches the rank stage). Cold is a secondary.

- **Gates and controls** are the k4c400 verifier's, with both sides on one base: one relation, probe count, scalar and index/layout digest set across BOTH binaries; rank work identical per seed across `kD4FO`, `kD4R1`, `kD4R1aa`; probe law in [0.80, 1.20] over at least 64 attempts for each binary; A/A end-to-end and cold point estimates inside [0.97, 1.03]; planted index-build lower bound above 1. An unknown gate fails.

Local checks (diagnostics, not timing evidence)

Both binaries reproduce D4's frozen identity (counts, index and layout digests, Q0 relation/probes/scalar, table digest) in 8 processes, with identical rank work across the binaries (`inputs/local_identity.json`, all checks pass). A smoke run of segments O1 and P1 through `run_schedule.py` and `arm.py` (10 rows, one per arm) was valid; it is not kept.

Files

`restore.py`, `make_log_table.py`, `check_log_table.py`, `local_identity.py`, `make_targets.py`, `check_panel_disjoint.py`, `check_seed_disjoint.py`, `make_power_prior.py`, `power_check.py`, `make_protocol.py`, `make_schedule.py`, `run_schedule.py`, `arm.py`, `verify_receipt.py`, `test_verify_receipt.py`, `band_model.py`; `protocol.json`, `schedule.json`; `inputs/`, `analysis/power.json`. Trigger: `.github/workflows/n83-fo2d4.yml`, on a push to `claude/n83-fo2d4`.